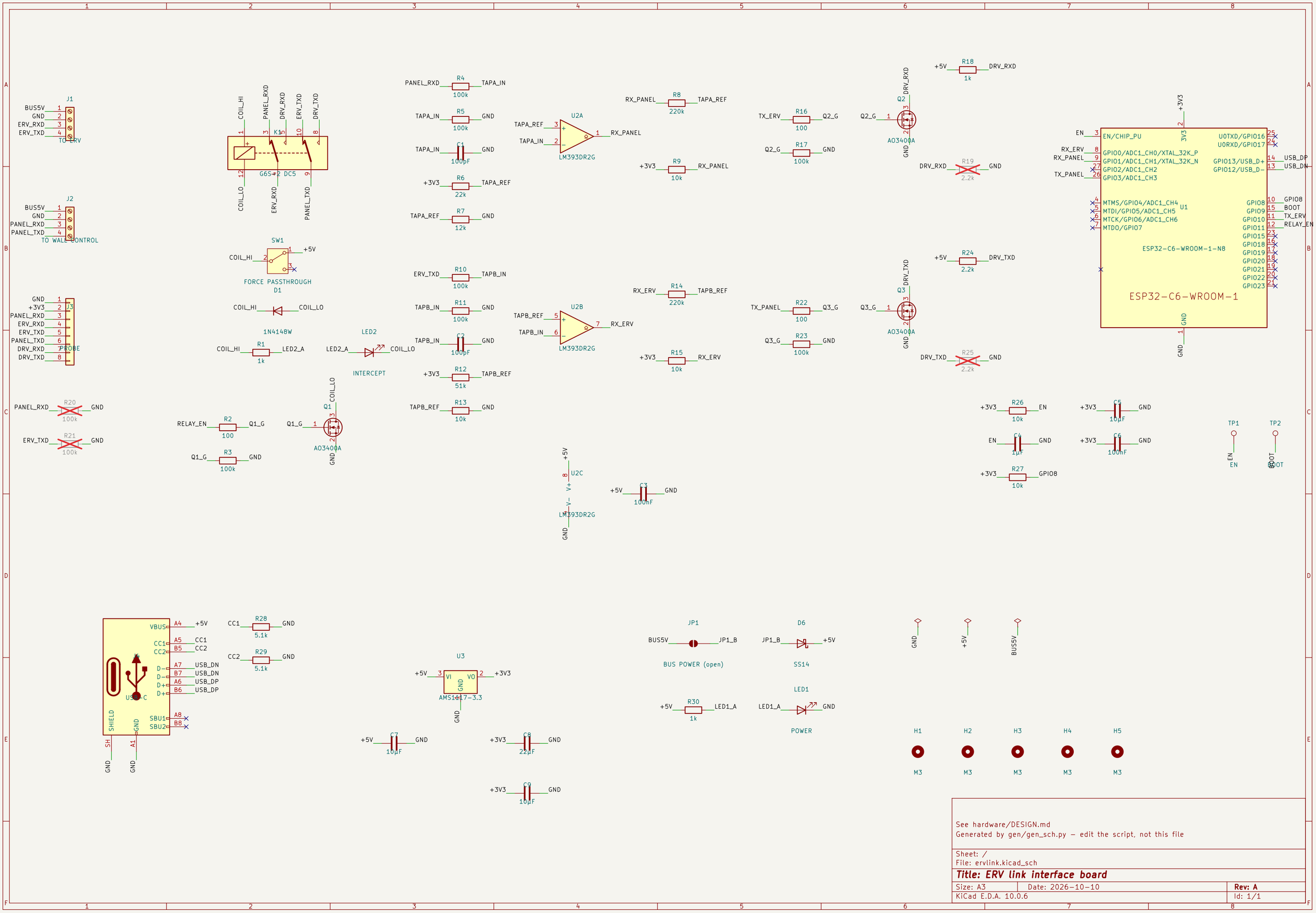

J1
BUS5V 1
GND 2
ERV_RXD 3
ERV_TXD 4
TO ERV
J2
BUS5V 1
GND 2
PANEL_RXD 3
PANEL_TXD 4
TO WALL CONTROL
J3
PROBE
GND 1
+3V3 2
PANEL_RXD 3
ERV_RXD 4
ERV_TXD 5
PANEL_TXD 6
DRV_RXD 7
DRV_TXD 8
K1
G6S-2 DC5
COIL_HI
PANEL_RXD
DRV_RXD
ERV_TXD
DRV_TXD
COIL_LO
ERV_RXD
PANEL_TXD
SW1
FORCE PASSTHROUGH
D1
1N4148W
LED2
INTERCEPT
R1 1k
R2 100
R3 100k
R4 100k
R5 100k
R6 22k
R7 12k
R8 220k
R9 10k
R10 100k
R11 100k
R12 51k
R13 10k
R14 220k
R15 10k
R16 100
R17 100k
R18 1k
R19 2.2k
R20 100k
R21 100k
R22 100
R23 100k
R24 2.2k
R25 2.2k
R26 10k
R27 10k
R28 5.1k
R29 5.1k
R30 1k
C1 100pF
C2 100pF
C3 100nF
C4 1µF
C5 10µF
C6 100nF
C7 10µF
C8 22µF
C9 10µF
U2A
U2B
U2C
LM393DR2G
Q1
Q2
Q3
AO3400A
U1
ESP32-C6-WROOM-1-N8
ESP32-C6-WROOM-1
EN/CHIP_PU
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
MTMS/GPIO4/ADC1_CH4
MTDI/GPIO5/ADC1_CH5
MTCK/GPIO6/ADC1_CH6
MTDO/GPIO7
U0TXD/GPIO16
U0RXD/GPIO17
GPIO13/USB_D+
GPIO12/USB_D-
GPIO8
GPIO9
GPIO10
GPIO11
GPIO15
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
3V3
GND
USB_DP
USB_DN
BOOT
TX_ERV
RELAY_EN
RX_ERV
RX_PANEL
TX_PANEL
U3
AMS1117-3.3
USB-C
VBUS
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
JP1
BUS POWER (open)
D6
SS14
LED1
POWER
TP1
TP2
H1
H2
H3
H4
H5
M3
See hardware/DESIGN.md
Generated by gen/gen_sch.py — edit the script, not this file
Sheet: /
File: ervlink.kicad_sch
Title: ERV link interface board
Size: A3
Date: 2026-10-10
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/1